

Chapter 1

Introduction

1.1. Background of the Study

Suicide is one of the most critical and under-addressed public health challenges in low- and middle-income countries (LMICs). Bangladesh, a densely populated South Asian nation, is no exception. Estimates suggest that the country witnesses approximately 10,000 suicide deaths annually, though the true figure is likely far greater due to widespread social stigma, cultural barriers, and the near-total absence of a systematic national suicide surveillance registry [1, 2]. Unlike high-income countries that maintain structured reporting mechanisms, suicide data in Bangladesh is predominantly captured through fragmented sources — newspaper reports, police records, and hospital admissions — each with their own inconsistencies and biases.

Among these sources, newspaper-reported data represents the richest and most accessible public record of suicide incidents in Bangladesh. Several researchers have relied on this source to understand patterns related to method, demographics, location, and motivating factors [3, 4]. However, the critical limitation of such datasets lies in their inherent incompleteness. Demographic details such as age, socioeconomic status, and occupation are frequently missing, while spatiotemporal and environmental attributes — latitude, longitude, weather conditions — are almost never directly recorded. The resulting missingness rates often exceed 50–90% for many important features, rendering conventional statistical and machine learning analyses unreliable or outright infeasible [5, 6].

This structural data gap creates a significant barrier for researchers and policymakers who wish to identify high-risk demographic groups, understand the environmental and social triggers of suicide, and design targeted prevention programmes. Addressing it requires a methodologically rigorous, reproducible pipeline that can handle severe missingness, preserve meaningful patterns, and ultimately surface actionable insights from incomplete real-world data.

1.2. Problem Statement

The absence of an official, structured suicide surveillance system in Bangladesh forces researchers to rely exclusively on media-reported cases. While such datasets are valuable as the primary available source, they are characterised by extreme and non-random miss-

ingness — often exceeding 50–90% for critical features such as socioeconomic indicators, geographic coordinates, and weather-related variables. Standard machine learning approaches applied directly to such incomplete data either fail entirely or produce heavily biased results [5,6].

Beyond the missingness problem, there is no established methodological guideline on which imputation technique performs best when combined with which clustering algorithm for this specific domain. In the absence of any ground-truth labels or external benchmarks, it is also unclear how to fairly evaluate and compare multiple unsupervised learning solutions — especially when cluster size imbalance and excessive noise make traditional validation metrics misleading [7].

Consequently, hidden patterns and risk profiles within existing newspaper-based suicide records remain undetected, directly hindering evidence-based suicide prevention efforts in Bangladesh.

1.3. Research Motivation

The primary motivation of this study is to bridge the gap between highly incomplete real-world suicide data and meaningful, interpretable pattern discovery. The research is driven by three interconnected observations:

First, Bangladesh lacks any algorithmic, data-driven analysis of its suicide landscape at scale. Existing studies have largely relied on manual grouping or simple statistical summaries, leaving a significant methodological void [3,4].

Second, the international literature on missing data imputation and unsupervised clustering — while extensive — has rarely been applied systematically to social and public health datasets from South Asia characterised by this level of missingness and mixed data types [7,8]. Existing benchmarks focus predominantly on clinical or genomic data, and no published work has applied a comprehensive, multi-method imputation–clustering pipeline to a large newspaper-derived social dataset from Bangladesh.

Third, standard clustering validation metrics are known to be misleading when clusters are severely imbalanced or when algorithms such as DBSCAN assign the majority of points as noise. A fairer, more interpretable evaluation framework is therefore needed for this application domain.

“Which combination of missing-data handling and unsupervised learning techniques produces the most reliable, interpretable, and actionable clusters from newspaper-based suicide records in Bangladesh?”

1.4. Research Objectives

The specific objectives of this research are:

To expand and enrich an existing publicly available dataset of 760 newspaper-reported suicide cases in Bangladesh (covering the year 2020) by manually collecting 857 additional cases spanning 2017–2025, and by introducing 9 new socioeconomic and clinical attributes, yielding a final dataset of 1,617 records and 34 analytical columns.

To perform exhaustive data cleaning, standardisation, and conditional filling of missing values using external geocoding (Geopy) and meteorological (Meteostat) APIs, substantially reducing missingness in spatial and weather-related features prior to any imputation.

To implement and systematically compare three state-of-the-art imputation methods — Multiple Imputation by Chained Equations (MICE) [9], K-Nearest Neighbours (KNN) Imputation [10], and MissForest [11] — each applied under two predictor-selection strategies (correlation-based context-aware and all-available no-context), producing six fully imputed datasets.

To apply four established clustering algorithms — K-Means [12], Agglomerative Clustering [13], DBSCAN [14], and Spectral Clustering [15] — across all eight datasets (six imputed and two non-imputed baselines), generating 32 complete imputation–clustering pipeline combinations.

To evaluate all 32 pipeline combinations using five established internal clustering validity indices — Silhouette Score, Calinski–Harabasz Index, Davies–Bouldin Index, Dunn Index, and Xie–Beni Index — supplemented by a newly proposed balance metric designed to penalise cluster size imbalance and excessive noise, combined into a single composite ranking score.

To identify the best-performing imputation–clustering combination and provide detailed interpretation of the resulting clusters across demographic, socioeconomic, geographic, and environmental dimensions, yielding actionable insights for suicide prevention in Bangladesh.

1.5. Scope and Significance of the Research

This research focuses on completed suicides reported in Bangladeshi newspapers between 2017 and 2025. It does not include suicide attempts, unreported cases, or data sourced from clinical or police records. The analytical scope is limited to unsupervised pattern discovery; no supervised prediction model is developed.

The significance of the research lies across three distinct areas:

Methodological contribution: The study provides a reproducible, end-to-end framework for handling severely incomplete, mixed-type social datasets through a systematic comparison of 32 imputation–clustering pipelines. The introduction of a balance-aware composite scoring mechanism represents a novel contribution to the clustering evaluation literature.

Empirical contribution: The study produces the first large-scale, algorithmically derived cluster analysis of newspaper-reported suicide data from Bangladesh, identifying distinct risk profiles that align with and extend existing sociological knowledge [3,4].

Societal contribution: The interpretable clusters and geographic risk patterns produced by the framework offer direct, evidence-based inputs for mental health policy-makers, NGOs, and public health professionals designing targeted suicide prevention programmes in Bangladesh.

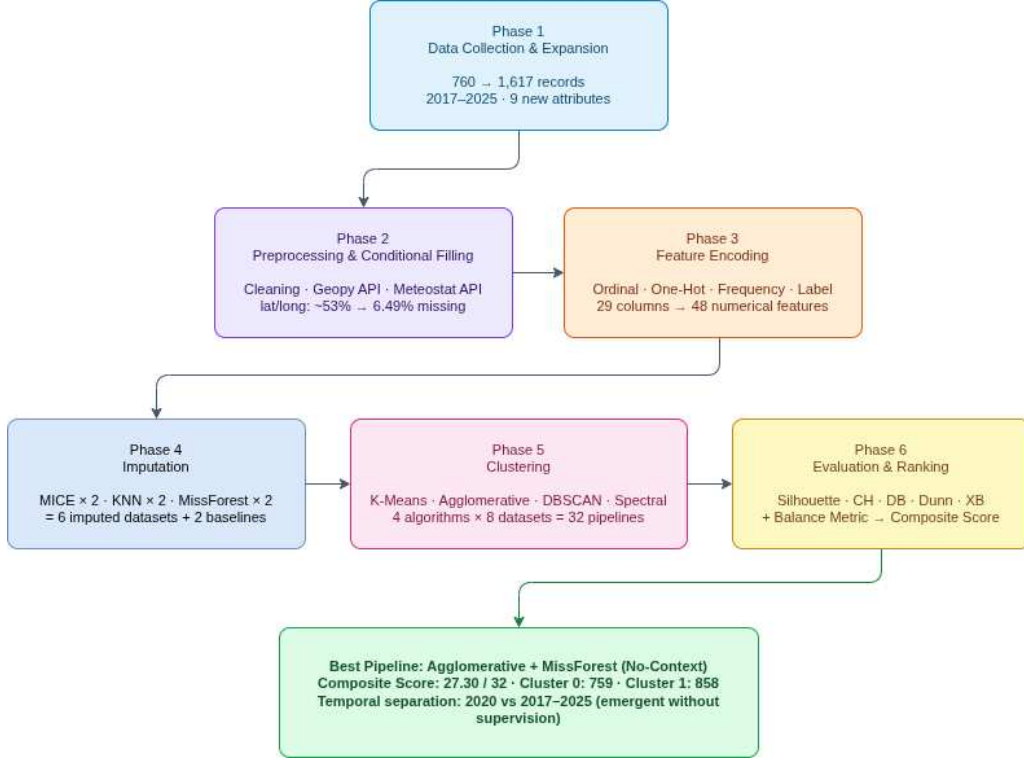


Figure 1.1: High-level overview of the six-phase research framework adopted in this study.

1.6. Research Contributions

The key contributions of this research are:

A significantly expanded, cleaned, and enriched suicide dataset comprising 1,617 records and 34 columns, made available for the broader research community and covering suicide incidents across Bangladesh from 2017 to 2025.

The first systematic benchmark comparing 32 imputation–clustering pipelines on real-world, severely incomplete Bangladeshi suicide data, covering three imputation methods, two predictor-selection strategies, and four clustering algorithms.

A novel balance-aware composite evaluation metric that integrates five standard internal validity indices with a cluster size equity penalty, providing a fairer ranking criterion for naturally imbalanced social datasets.

Actionable demographic, environmental, and geographic risk profiles derived from the best-performing pipeline (Agglomerative Clustering with MissForest no-context imputation), directly interpretable for prevention stakeholders.

A demonstration that carefully designed imputation and clustering pipelines can recover temporally coherent, structurally meaningful patterns from data with over 90% missingness in several features — without any supervised labelling or domain-specific constraints imposed during modelling.

1.7. Thesis Organisation

The remainder of this report is organised as follows:

Chapter 2 presents a background study and literature review covering existing research on suicide in Bangladesh, the challenges of newspaper-based datasets, and related work on missing data imputation and clustering in public health contexts.

Chapter 3 describes the detailed research methodology, including the data collection and expansion process, the preprocessing and encoding pipeline, the three imputation strategies and their variants, the four clustering algorithms, and the composite evaluation framework.

Chapter 4 discusses the implementation environment, experimental results across all 32 pipelines, group-level comparisons, and a detailed cluster-level profiling of the best-performing pipeline.

Chapter 5 addresses engineering standards, societal impacts, ethical considerations, project challenges and constraints, and the project timeline.

Chapter 6 concludes the thesis with a summary of findings, achievement of research objectives, limitations of the study, and directions for future work.